

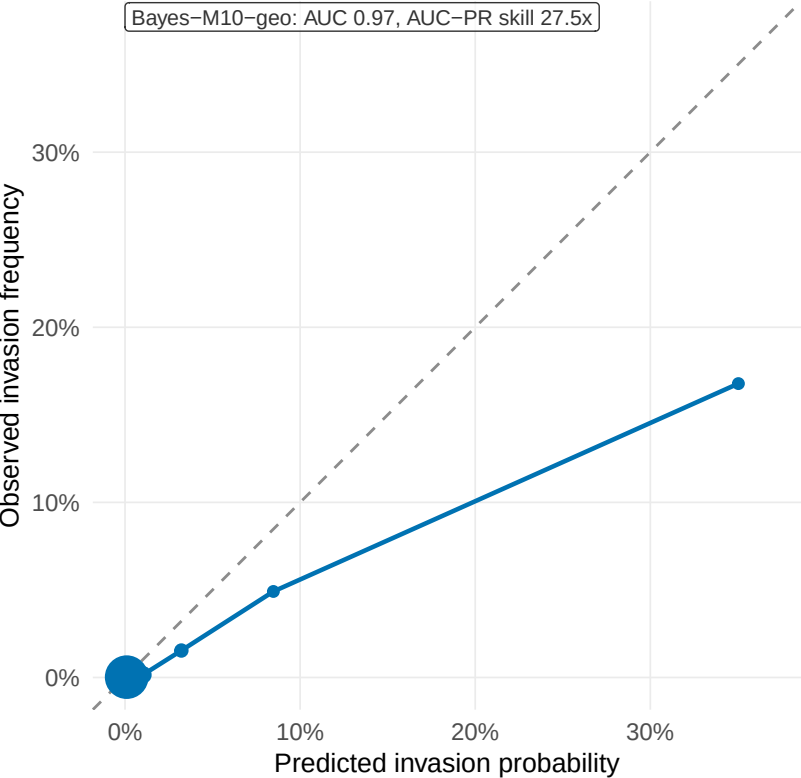
# How good is the invasion model? (h = 2w)

(A) calibration + AUC and (B) real-time prioritisation vs random – after Kraemer & Cauchemez 2017, Lancet Inf Dis, Fig 3

## (A) Calibration: predicted vs observed invasion (h = 2w)

On the diagonal = calibrated probabilities; below = over-confident.

zone-weeks   ● 1000   ● 2000   ● 3000   ● 4000   ● 5000   —●— Bayes-M10-geo

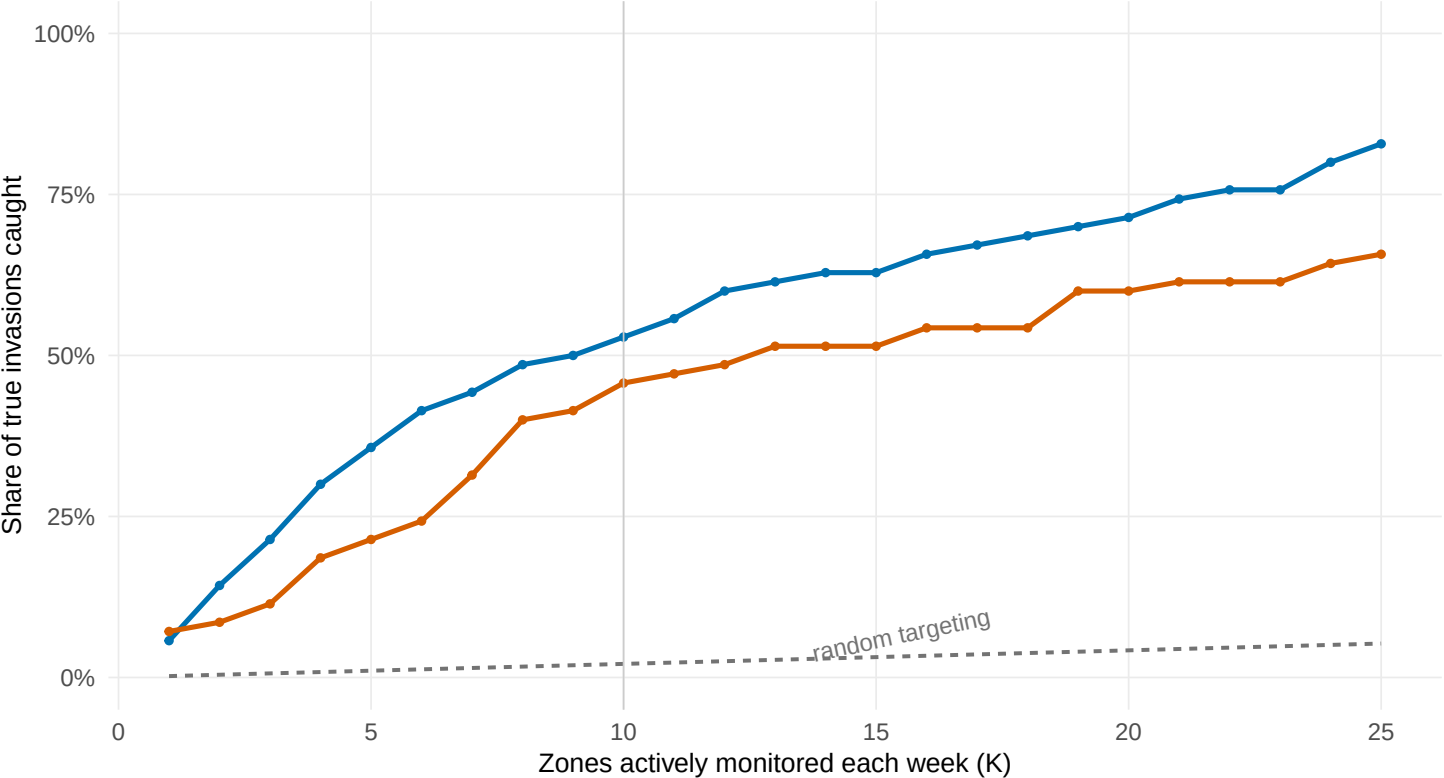


The 4.0% of at-risk zone-weeks we score >15%/week had a 17% observed invasion rate.

## (B) Prioritisation: invasions caught vs random targeting

Share of true next-week invasions caught at a fixed weekly alert budget of K zones, pooled over LFO folds

—●— Bayes-M10-geo   —●— Naive-epicentre-inflow



At a budget of K=10 zones/week, Bayes-M10-geo catches 53% of the next invasions vs 2% for random targeting (25.1x better).